



REACT DEVELOPMENT

Complete Learning Roadmap

Master component architecture, state orchestration, and performance for complex React apps.

JSX mental model

Component composition

Props and state

Event handling

Hooks patterns

Controlled forms

01

Fundamentals

02

Core Concepts

03

Tools and
Frameworks

04

Real Projects

05

Advanced Topics

5

Learning Stages

20+

Core Topics

10+

Projects

100%

React

Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

JSX mental model •
Component composition •
Props and state • Event
handling

02

Core Concepts

Hooks patterns • Controlled
forms • Context boundaries •
Error boundaries

03

Tools and Frameworks

React Router

04

Real Projects

Admin console build •
Realtime feed UI • Multi-step
workflow app • Accessible
component library

05

Advanced Topics

Render optimization •
Concurrent features • Large
app architecture •
Performance profiling

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- JSX mental model
- Component composition
- Props and state
- Event handling

Tools & Frameworks

- JSX mental model
- Component composition
- Props and state
- Event handling

Recommended Projects

- React Development workshop using JSX mental model
- React Development portfolio project with Component composition

Stage 1 Milestone Complete React Development workshop using JSX mental model and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Hooks patterns
- Controlled forms
- Context boundaries
- Error boundaries

Tools & Frameworks

- Hooks patterns
- Controlled forms
- Context boundaries
- Error boundaries

Recommended Projects

- React Development workshop using Hooks patterns
- React Development portfolio project with Controlled forms

Stage 2 Milestone Complete React Development workshop using Hooks patterns and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- React Router
- State libraries
- Data-fetching libraries
- Testing libraries

Tools & Frameworks

- React Router

Recommended Projects

- React Development workshop using React Router
- React Development portfolio project with State libraries

Stage 3 Milestone Complete React Development workshop using React Router and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Admin console build
- Realtime feed UI
- Multi-step workflow app
- Accessible component library

Tools & Frameworks

- Admin console build
- Realtime feed UI
- Multi-step workflow app
- Accessible component library

Recommended Projects

- React Development workshop using Admin console build
- React Development portfolio project with Realtime feed UI

Stage 4 Milestone Complete React Development workshop using Admin console build and document the decisions, trade-offs, and validation checks before moving to Stage 5.



Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

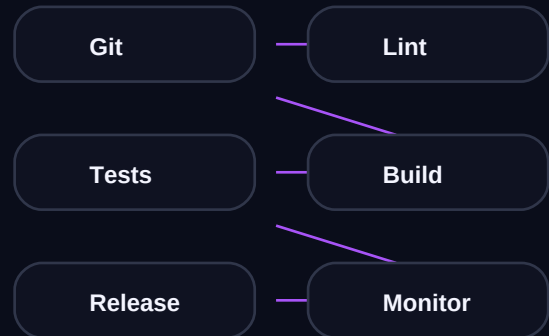
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Render optimization
- Concurrent features
- Large app architecture
- Performance profiling

CI/CD Pipeline



Stage 5 Milestone

A production-grade react development workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

React Development docs and community guides

Link: <https://react.dev/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/react>

PAID

Structured specialization track focused on React Development

Link: <https://roadmap.sh/react>

FREE

React Development roadmap reference

Link: <https://roadmap.sh/react>

FREE

Official documentation

Link: <https://react.dev/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in React Development?
- Which project best proves Stage 4 competency in React Development?
- What trade-offs appear in Stage 5 decisions for React Development?